

Paper 1

1. Time taken for X to fall: $t = \frac{v \sin \theta}{g}$. Time taken for Y to fall: $t = \sqrt{\frac{2h}{g}}$.

a) Passing by the window: $s = ut + \frac{1}{2}at^2$

Velocity at top of window: $u = \frac{h_w - \frac{1}{2}gt_w^2}{t_w} = 28.57791471 \text{ m/s}$

Falling down: $u = \sqrt{2g(h - (50 + 5))}$

$h = \frac{u^2}{2g} + 55 = 96.62574969 \approx \mathbf{97m}$

b) $\frac{v \sin \theta}{g} = \sqrt{\frac{2h}{g}}$

$v = \sqrt{\frac{2gh}{\sin^2 \theta}} = 87.08150685 \approx \mathbf{87 \text{ m/s}}$

c) $R = \frac{v^2 \sin 2\theta}{g} = 669.4428311 \approx \mathbf{670m}$

2. The object slides when friction is no longer enough to accelerate the object to follow the acceleration of the platform, which is $a_{max} = \omega^2 A$

a) $f = \mu mg = ma$. $\mu mg = m\omega^2 A$

$$\mu = \frac{\omega^2 A}{g} = \frac{\left(\frac{2\pi}{T}\right)^2 A}{g} = 0.4024303527 \approx \mathbf{0.40}$$

- b) The maximum acceleration must be equal to -g at the highest point.

$$\omega^2 A = g$$

$$A = \frac{g}{\left(\frac{2\pi}{T}\right)^2} = 0.5591529565 \approx \mathbf{0.56m}$$

3. Based on mark allocation, we shift some working to part b).

a) $\Delta P = -\rho g \Delta z$

b) $PV = NkT$. $\rho = \frac{M}{V} = \frac{Nm}{V} = \frac{Pm}{kT}$. Hence $\Delta P = -\frac{Pm}{kT} g \Delta z$

$$dP = -\frac{mg}{kT} \cdot P \cdot dz$$

$$\int_{P(0)}^{P(z)} \frac{1}{P} dP = \int_0^z -\frac{mg}{kT} dz$$

$$[\ln P]_{P(0)}^{P(z)} = -\frac{mgz}{kT}$$

$$P(z) = P(0)e^{-\frac{mgz}{kT}}$$

4. I have to write something here otherwise the tabbing doesn't work

$$a) \quad v = \sqrt{\frac{F_T}{\mu}} = \sqrt{\frac{F_T}{\frac{m}{l}}} = \sqrt{\frac{F_T \cdot l}{m}}$$

b) The copper has a higher μ than brass, because zinc is lighter. Its velocity and hence frequency of the wave is lower. Thus its frequency is $256 - 5 = 251 \text{ Hz}$. For fundamental mode, $\lambda = 2l$.

$$v = f\lambda = \sqrt{\frac{F_T \cdot l}{m}}$$

$$f_c \cdot 2l = \sqrt{\frac{F_T \cdot l}{m_c}}$$

$$4f_c^2 l m_c = F_T$$

Similarly, $4f_b^2 l m_b = F_T$. Hence $f_c^2 m_c = f_b^2 m_b$.

$$\frac{m_b}{m_c} = \frac{f_c^2}{f_b^2}$$

Let the proportion by length of zinc be x . Then

$$\frac{x \cdot \rho_z + (1 - x) \cdot \rho_c}{1 \cdot \rho_c} = \frac{m_b}{m_c} = \frac{f_c^2}{f_b^2}$$

$$\rho_c - (\rho_c - \rho_z)x = \frac{f_c^2}{f_b^2} \cdot \rho_c$$

$$x = \frac{\left(1 - \frac{f_c^2}{f_b^2}\right)}{\rho_c - \rho_z} \cdot \rho_c = 0.1921157837$$

Then the proportion by mass of zinc is:

$$\frac{x \cdot \rho_z}{x \cdot \rho_z + (1 - x) \cdot \rho_c} = 0.1596085776 = \mathbf{15.96\%}$$

5. I have to write something here otherwise the tabbing doesn't work

$$a) \quad k = \frac{\ln 2}{T_{\frac{1}{2}}} = \mathbf{5.998158 \times 10^{-5} \text{ min}^{-1}}$$
 (you could use days as well)

$$b) \quad N = N_0 e^{-kt}, \dot{N} = -kN_0 e^{-kt}.$$

Decay rate in the sample: $\frac{3171}{30} \text{ min}^{-1}$. Decay rate in the body: $\frac{3171}{30} \cdot \frac{V}{5 \text{ ml}}$

$$\text{Number of iodine: } N_0 = \frac{\text{total mass}}{\text{individual mass}} = \frac{5 \times 10^{-6} \times 1.00 \times 10^{-10}}{131 \times 1.66 \times 10^{-27}}$$

$$\frac{3171}{30} \cdot \frac{V}{5 \text{ ml}} = kN_0 e^{-kt}$$

$$V = 5 \text{ ml} \cdot \frac{30}{3171} \cdot kN_0 e^{-kt} = 5984.006266 \text{ ml} \approx \mathbf{5.98 \text{ L}}$$

Paper 2

1. Borrowing method from relativistic 4 vector, but in regular momentum:

a) $\vec{p}_1 + \vec{p}_2 = \vec{p}_3 + \vec{p}_4$. But actually, $\vec{p}_2 = 0$ for the ionized He.

$$|\vec{p}_1 - \vec{p}_3|^2 = |\vec{p}_4|^2$$

$$p_1^2 + p_3^2 - 2\vec{p}_1 \cdot \vec{p}_3 = p_4^2$$

But actually $\vec{p}_1 \cdot \vec{p}_3 = 0$ as $\vec{p}_1 \perp \vec{p}_3$. Using $p^2 = 2mE$ for normal objects:

$$2m_n E_1 + 2m_n E_3 = 2m_\alpha E_4$$

$$E_1 + E_3 = 4E_4$$

Now considering the conservation of energy, $E_1 = E_3 + E_4 + E_{excite}$, where $E_{excite} = 54.4 \times \left(1 - \frac{1}{n^2}\right) eV$. Hereby referred to as E_x . Solving the simultaneous equation, we have:

$$E_3 = \frac{1}{5}(3E_1 - 4E_x) = \frac{43.52}{n^2} - 2.6 eV$$

$$E_4 = \frac{1}{5}(2E_1 - E_x) = \frac{10.88}{n^2} + 16.4 eV$$

b) $54.4 \left(1 - \frac{1}{n^2}\right) = \frac{hc}{\lambda}$. Observing the equation for E_3 , we are limited to $n^2 \leq \frac{43.52}{2.6} = 16.7, n \leq 4$.

$$\lambda = \frac{hc}{54.4 \left(1 - \frac{1}{n^2}\right)}$$

Shortest from $n = 4 \rightarrow n = 1$

$$\lambda = \frac{hc}{54.4 \left(1 - \frac{1}{4^2}\right)} = 2.4375 \times 10^{-8} m$$

Longest from $n = 4 \rightarrow n = 3$:

$$\lambda = \frac{hc}{54.4 \left(\frac{1}{3^2} - \frac{1}{4^2}\right)} = 4.70089286 \times 10^{-7} m$$

c) Let's go into the frame of the CM. Then let the photon go forwards and the atom go backwards, with momentum $\frac{h}{\lambda} = p$. By conservation of energy: $\Delta E = \frac{p^2}{2m} + \frac{hc}{\lambda} = \frac{p^2}{2m} + pc$. By quadratic formula:

$$p^2 + 2mc \cdot p - 2m\Delta E = 0$$

$$p = \frac{-2mc \pm \sqrt{4m^2c^2 + 8m\Delta E}}{2}$$

$$p = -mc + \sqrt{m^2c^2 + 2m\Delta E} = 2.71999998 \times 10^{-26}$$

The energy wasted is then $\Delta E - pc$, percentage is $\% = 1 - \frac{pc}{\Delta E}$, where $\Delta E = 54.4 \times \left(1 - \frac{1}{4^2}\right)$. We get **7.35294200** $\times 10^{-9}$

d) Simply $\Delta v = \frac{p}{m} = 4.09638551 m/s$

2. Terminal velocity is when it's balanced:

a) $kv = mg$

$$v = \frac{mg}{k} = 90 \text{ m/s}$$

b) $m\ddot{x} = -mg - k\dot{x}$

$$\ddot{x} = -\frac{k}{m}\dot{x} - g$$

Using $\ddot{x} = \frac{d\dot{x}}{dt}$, we have $\frac{1}{\frac{k}{m}\dot{x} + g} d\dot{x} = -dt$

$$\int_0^v \frac{1}{\frac{k}{m}\dot{x} + g} d\dot{x} = -\int_0^t dt$$

$$\left[\frac{m}{k} \ln \left| \frac{k}{m}\dot{x} + g \right| \right]_0^v = -t$$

$$\frac{m}{k} \ln \left| \frac{kv}{mg} + 1 \right| = -t$$

$$\frac{kv}{mg} + 1 = e^{-\frac{kt}{m}}$$

$$v = \frac{mg}{k} \left(e^{-\frac{kt}{m}} - 1 \right)$$

Note that velocity is negative here as it is downwards. At 99%,

$$0.99 = \left| e^{-\frac{kt}{m}} - 1 \right|$$

$$e^{-\frac{kt}{m}} = 0.01$$

$$t = -\frac{m}{k} \ln 0.01 = \frac{m}{k} \ln 100 = 42.2492678 \text{ s}$$

c) $\frac{dx}{dt} = \frac{mg}{k} \left(e^{-\frac{kt}{m}} - 1 \right)$

$$\int_H^x \frac{k}{mg} dx = \int_0^t \left(e^{-\frac{kt}{m}} - 1 \right) dt$$

$$\frac{k}{mg} (x - H) = \left[-\frac{m}{k} e^{-\frac{kt}{m}} - t \right]_0^t$$

$$\frac{k}{mg} (H - x) = \frac{m}{k} \left(e^{-\frac{kt}{m}} - 1 \right) + t$$

$$(H - x) = \frac{m^2 g}{k^2} \left(e^{-\frac{kt}{m}} - 1 \right) + \frac{mg}{k} t = 2985.00291 \text{ m}$$

3. Lorentz force $F = q\vec{v} \times \vec{B}$

a) $qvB = mv\omega$

$$\frac{qB}{m} = \frac{2\pi}{T}$$

$$T = \frac{2\pi m}{qB}$$

b) $F = q\vec{v} \times \vec{B} = q \begin{pmatrix} v_x \\ v_y \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ B_z \end{pmatrix} = q \begin{vmatrix} i & j & k \\ v_x & v_y & 0 \\ 0 & 0 & B_z \end{vmatrix} = \begin{pmatrix} qv_y B_z \\ -qv_x B_z \\ 0 \end{pmatrix}$

Since $F = ma$, we have $\vec{a} = \begin{pmatrix} \frac{qv_y B_z}{m} \\ -\frac{qv_x B_z}{m} \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{2\pi v_y}{T} \\ -\frac{2\pi v_x}{T} \\ 0 \end{pmatrix}$

c) $x(t + \Delta t) = x(t) + v_x(t)\Delta t + \frac{1}{2}a_x(t)\Delta t^2 = x(t) + v_x(t)\Delta t + \frac{\pi v_y}{T}\Delta t^2$
 $y(t + \Delta t) = y(t) + v_y(t)\Delta t + \frac{1}{2}a_y(t)\Delta t^2 = y(t) + v_y(t)\Delta t - \frac{\pi v_x}{T}\Delta t^2$

d) Square both equations for v_x, v_y , since $v_x^2 + v_y^2 = v^2$

$$v_{x\text{ new}}^2 = v_x^2 + \frac{2\pi}{T}v_x v_y \Delta t + \frac{4\pi^2}{T^2}v_y^2 \Delta t^2$$

$$v_{y\text{ new}}^2 = v_y^2 - \frac{2\pi}{T}v_x v_y \Delta t + \frac{4\pi^2}{T^2}v_x^2 \Delta t^2$$

$$v_{\text{new}}^2 = v_x^2 + v_y^2 + \frac{4\pi^2}{T^2}(v_x^2 + v_y^2)\Delta t^2$$

$$v_{\text{new}}^2 = v^2 \left(1 + \frac{4\pi^2}{T^2}\Delta t^2 \right)$$

$$\Delta E_k = E_k \times \frac{4\pi^2}{T^2}\Delta t^2 = 4\lambda^2 E_k$$

Alternatively, every time, energy is added in the form of $a\Delta t$ perpendicular to the current velocity. The acceleration is not perpendicular to velocity during the time step.

e) $\frac{E'}{E_0} = \left(1 + \frac{4\pi^2}{T^2}\Delta t^2 \right)^{\frac{T}{\Delta t}}$, note that $\frac{T}{\Delta t}$ is the number of iterations required for one complete circle. Then $\frac{E'}{E_0} = (1 + 4\pi^2 \times 0.01^2)^{\frac{1}{0.01}} = 1.48291085$

- f) $a_x = \frac{x(t+\Delta t) + x(t-\Delta t) - 2x(t)}{\Delta t^2} = \frac{2\pi v_y}{T} = \frac{2\pi}{T} \left(\frac{y(t+\Delta t) - y(t-\Delta t)}{2\Delta t} \right)$. Here, the special definition of velocity is used.

$$x(t + \Delta t) + x(t - \Delta t) - 2x(t) = \frac{\pi}{T} \Delta t (y(t + \Delta t) - y(t - \Delta t))$$

$$x(t + \Delta t) - \frac{\pi \Delta t}{T} y(t + \Delta t) = -\frac{\pi \Delta t}{T} y(t - \Delta t) - x(t - \Delta t) + 2x(t)$$

Similarly,

$$y(t + \Delta t) + y(t - \Delta t) - 2y(t) = -\frac{\pi}{T} \Delta t (x(t + \Delta t) - x(t - \Delta t))$$

$$y(t + \Delta t) + \frac{\pi \Delta t}{T} x(t + \Delta t) = \frac{\pi \Delta t}{T} x(t - \Delta t) - y(t - \Delta t) + 2y(t)$$

Solving the simultaneous equation:

$$x_+ = \frac{1}{1 + \lambda^2} [-2\lambda y_- + (\lambda^2 - 1)x_- + 2\lambda y + 2x]$$

$$y_+ = \frac{1}{1 + \lambda^2} [2\lambda x_- + (\lambda^2 - 1)y_- + 2y - 2\lambda x]$$

Where $\lambda = \frac{\pi \Delta t}{T} = \frac{qB\Delta t}{2m}$.

- g) $v_x = \frac{x_+ - x_-}{2\Delta t} = \frac{1}{2\Delta t(1 + \lambda^2)} [-2\lambda y_- - 2x_- + 2\lambda y + 2x] = \frac{(x - x_-) + \lambda(y - y_-)}{\Delta t(1 + \lambda^2)}$
 $v_y = \frac{y_+ - y_-}{2\Delta t} = \frac{1}{2\Delta t(1 + \lambda^2)} [2\lambda x_- - 2y_- + 2y - 2\lambda x] = \frac{(y - y_-) - \lambda(x - x_-)}{\Delta t(1 + \lambda^2)}$

Now set $t \rightarrow t + \Delta t$

$$v_{x+} = \frac{(x_+ - x) + \lambda(y_+ - y)}{\Delta t(1 + \lambda^2)}$$

$$v_{y+} = \frac{(y_+ - y) - \lambda(x_+ - x)}{\Delta t(1 + \lambda^2)}$$

$$v_{x+} + v_x = \frac{(x_+ - x_-) + \lambda(y_+ - y_-)}{\Delta t(1 + \lambda^2)} = \frac{2v_x + 2\lambda v_y}{(1 + \lambda^2)}$$

$$v_{y+} + v_y = \frac{(y_+ - y_-) - \lambda(x_+ - x_-)}{\Delta t(1 + \lambda^2)} = \frac{2v_y - 2\lambda v_x}{(1 + \lambda^2)}$$

$$v_{x+} = \frac{(1 - \lambda^2)v_x + 2\lambda v_y}{(1 + \lambda^2)}; \quad v_{y+} = \frac{(1 - \lambda^2)v_y - 2\lambda v_x}{(1 + \lambda^2)}$$

$$v_{x+}^2 + v_{y+}^2 = \frac{1}{(1 + \lambda^2)^2} \left[\left((1 - \lambda^2)v_x + 2\lambda v_y \right)^2 + \left((1 - \lambda^2)v_y - 2\lambda v_x \right)^2 \right]$$

$$v_{x+}^2 + v_{y+}^2 = \frac{1}{(1 + \lambda^2)^2} \left[[(1 - \lambda^2)^2 + 4\lambda^2](v_x^2 + v_y^2) \right] = v^2$$

$$v_{new}^2 = v^2$$

$$\Delta E = 0$$

This is called the Boris Method, and it is a proper method in research, so we are not surprised that it has no error in energy (in fact to the second order as well). <http://www.unige.ch/~hairer/preprints/borisbea.pdf>

4. The pulses take the same time to reach the spaceship as they take to come back, in the earth's frame.

a) $x = \frac{5 \times 60 \times c}{2} = 150c = 4.5 \times 10^9 \text{ m}$

- b) The light travelled for an extra $\frac{\Delta t}{2}$ seconds, that is the contracted length of the spaceship + the movement of the spaceship

$$\frac{c\Delta t}{2} = \frac{L}{\gamma} + \frac{v\Delta t}{2}$$

$$\frac{c\Delta t}{2} = L\sqrt{1-\beta^2} + \frac{\beta c\Delta t}{2}$$

$$\frac{c\Delta t}{2}(1-\beta) = L\sqrt{1-\beta^2}$$

$$\frac{c\Delta t}{2L} = \sqrt{\frac{1+\beta}{1-\beta}}$$

$$\left(\frac{c\Delta t}{2L}\right)^2 = \frac{1+\beta}{1-\beta} = \frac{2}{1-\beta} - 1$$

$$\left(\frac{c\Delta t}{2L}\right)^2 + 1 = \frac{2}{1-\beta}$$

$$1-\beta = \frac{2}{\left(\frac{c\Delta t}{2L}\right)^2 + 1}$$

$$\beta = 1 - \frac{2}{\left(\frac{c\Delta t}{2L}\right)^2 + 1} = \frac{4}{5}, \text{ so } v = 0.8c$$

- c) Recall $\begin{pmatrix} x' \\ ict' \end{pmatrix} = \begin{pmatrix} \gamma & i\gamma\beta \\ -i\gamma\beta & \gamma \end{pmatrix} \begin{pmatrix} x \\ ict \end{pmatrix}$, so $ct' = -\gamma\beta x + \gamma ct$

Event 1: $\begin{pmatrix} x \\ ct \end{pmatrix} = \begin{pmatrix} 150c \\ 150c \end{pmatrix}$

Event 2: $\begin{pmatrix} x \\ ct \end{pmatrix} = \begin{pmatrix} \left(150 + \frac{\Delta t}{2}\right)c \\ \left(150 + \frac{\Delta t}{2}\right)c \end{pmatrix}$

Event 1': $ct' = -\gamma\beta \cdot 150c + \gamma c \cdot 150 = 150\gamma c(1-\beta)$

Event 2': $ct' = -\gamma\beta \cdot 150c + \gamma c \cdot 150 = \left(150 + \frac{\Delta t}{2}\right)\gamma c(1-\beta)$

$$\Delta t' = \frac{\Delta t}{2}\gamma(1-\beta) = 2\mu s$$

Proof of Length Contraction: the two events in S' frame should be distance L apart. Also, $x' = ct'$ as the speed of light is constant. Thus $\left(\frac{\Delta t}{2}\right)\gamma c(1-\beta) = L$, which is $\frac{c\Delta t}{2}(1-\beta) = \frac{L}{\gamma}$, the same result as above.

- d) Normal time dilation: $\Delta t = \gamma\Delta t'$

- They are related by $\gamma(1-\beta)$ instead of simply γ , because in the simple time dilation case, time is transferred "instantaneously", but here, the time is measured by the transmission of light to and from the spaceship. This causes a time lag.
- Time is dilated from "proper time", which is when two events happen at the same place in a frame. Here, the two events are not at the same place, thus the $\Delta t'$ is not proper time.